

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	N. Vinay Yadav	Reg. No.:	192472061
Section / Batch:	CSE – AI / 3 rd Year	Date:	09-10-2026

Code of Conduct

I N. Vinay Yadav (192472061) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Information Extraction	Regex-Based Information Extraction	1
Pattern Matching	Regex Pattern Matching	2
Regular Expression Patterns	Regex-Based Input Validation	3

Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.

Identify email addresses.

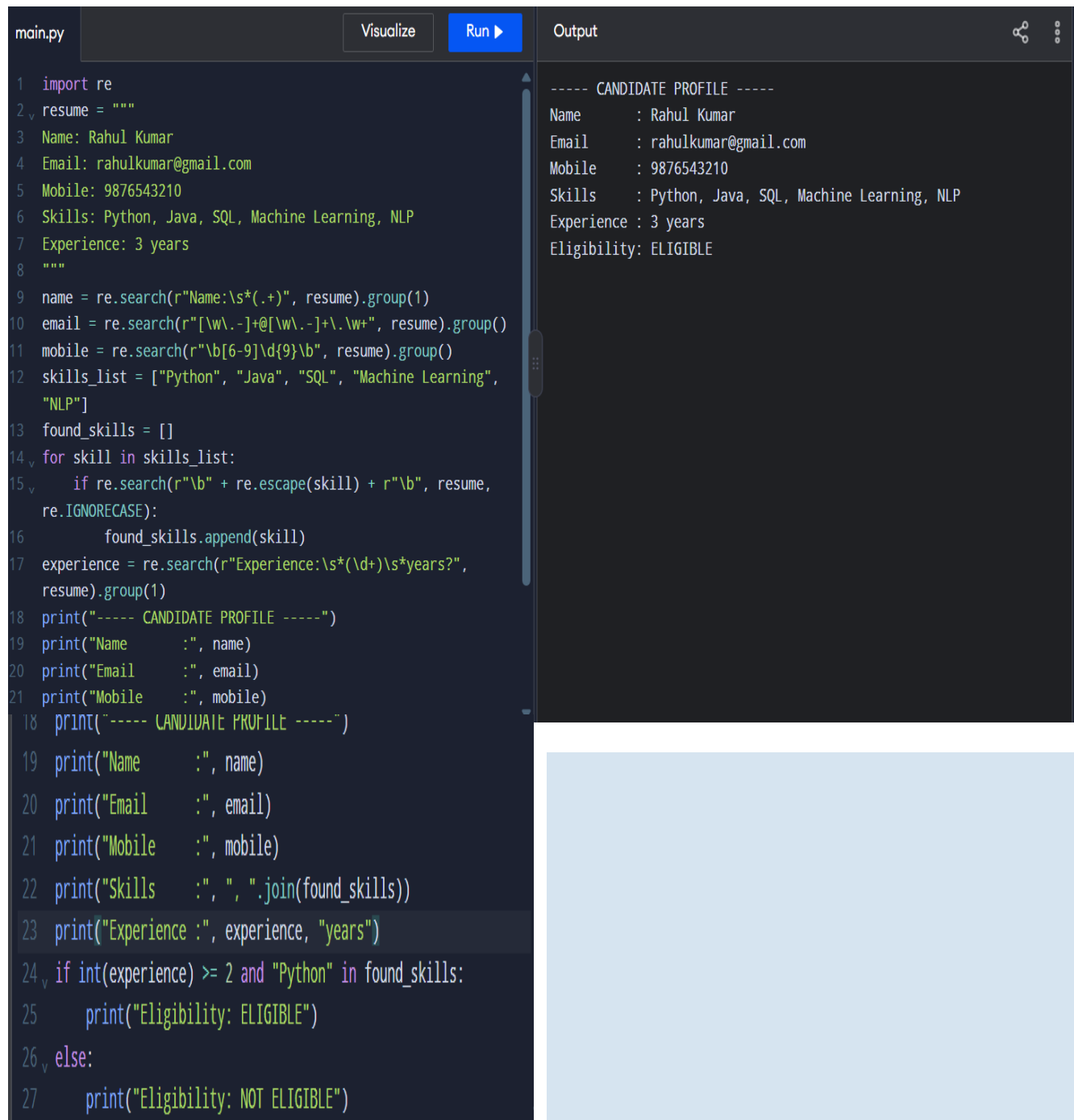
Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP).

Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).



```
main.py Visualize Run ▶ Output
1 import re
2 resume = """
3 Name: Rahul Kumar
4 Email: rahulkumar@gmail.com
5 Mobile: 9876543210
6 Skills: Python, Java, SQL, Machine Learning, NLP
7 Experience: 3 years
8 """
9 name = re.search(r"Name:\s*(.+)", resume).group(1)
10 email = re.search(r"[w\.-]+@[w\.-]+\w+", resume).group()
11 mobile = re.search(r"\b[6-9]\d{9}\b", resume).group()
12 skills_list = ["Python", "Java", "SQL", "Machine Learning",
13               "NLP"]
14 found_skills = []
15 for skill in skills_list:
16     if re.search(r"\b" + re.escape(skill) + r"\b", resume,
17                 re.IGNORECASE):
18         found_skills.append(skill)
19 experience = re.search(r"Experience:\s*(\d+)\s*years?",
20                       resume).group(1)
21 print("----- CANDIDATE PROFILE -----")
22 print("Name      :", name)
23 print("Email      :", email)
24 print("Mobile     :", mobile)
25 print("Skills      :", ", ".join(found_skills))
26 print("Experience :", experience, "years")
27 if int(experience) >= 2 and "Python" in found_skills:
28     print("Eligibility: ELIGIBLE")
29 else:
30     print("Eligibility: NOT ELIGIBLE")
```

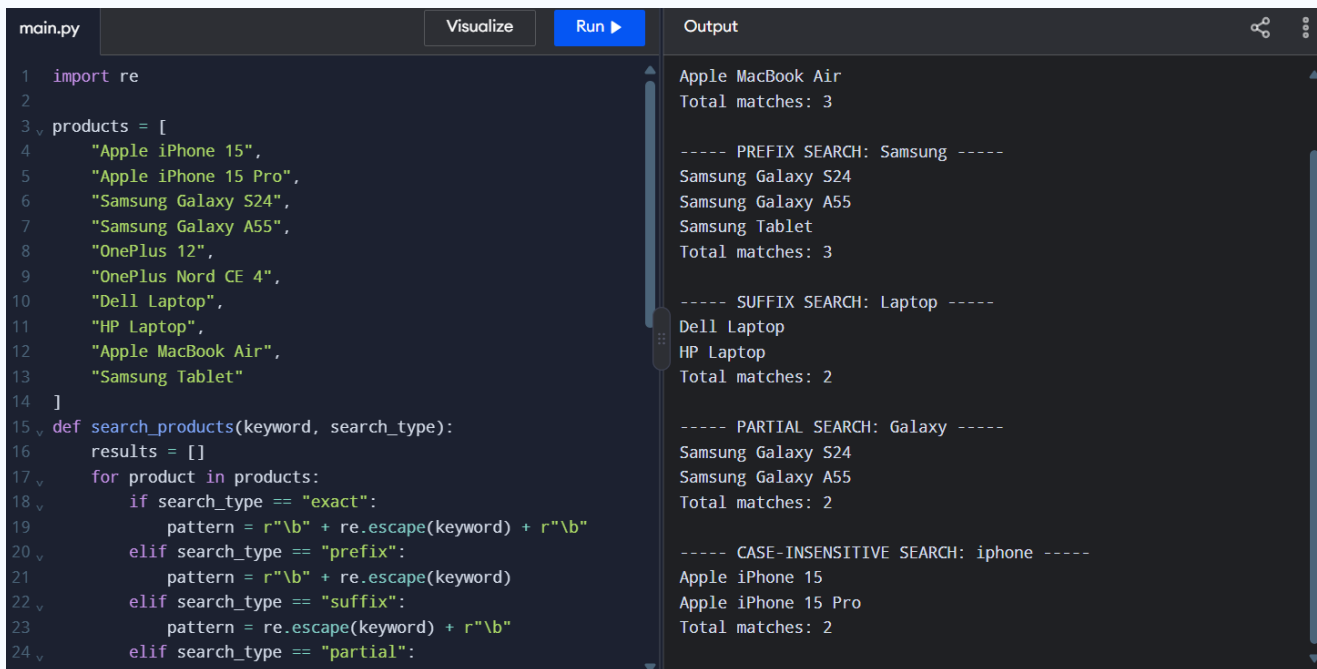
----- CANDIDATE PROFILE -----
Name : Rahul Kumar
Email : rahulkumar@gmail.com
Mobile : 9876543210
Skills : Python, Java, SQL, Machine Learning, NLP
Experience : 3 years
Eligibility: ELIGIBLE

Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.



The screenshot shows a Python IDE with a file named `main.py`. The code defines a list of products and a function `search_products` that uses regular expressions to search for products based on a keyword and a search type. The output window shows the results of several searches.

```
main.py Visualize Run ▶
```

```
1 import re
2
3 products = [
4     "Apple iPhone 15",
5     "Apple iPhone 15 Pro",
6     "Samsung Galaxy S24",
7     "Samsung Galaxy A55",
8     "OnePlus 12",
9     "OnePlus Nord CE 4",
10    "Dell Laptop",
11    "HP Laptop",
12    "Apple MacBook Air",
13    "Samsung Tablet"
14 ]
15 def search_products(keyword, search_type):
16     results = []
17     for product in products:
18         if search_type == "exact":
19             pattern = r"^\b" + re.escape(keyword) + r"\b$"
20         elif search_type == "prefix":
21             pattern = r"^\b" + re.escape(keyword)
22         elif search_type == "suffix":
23             pattern = re.escape(keyword) + r"\b$"
24         elif search_type == "partial":
25             pattern = re.escape(keyword)
```

Output

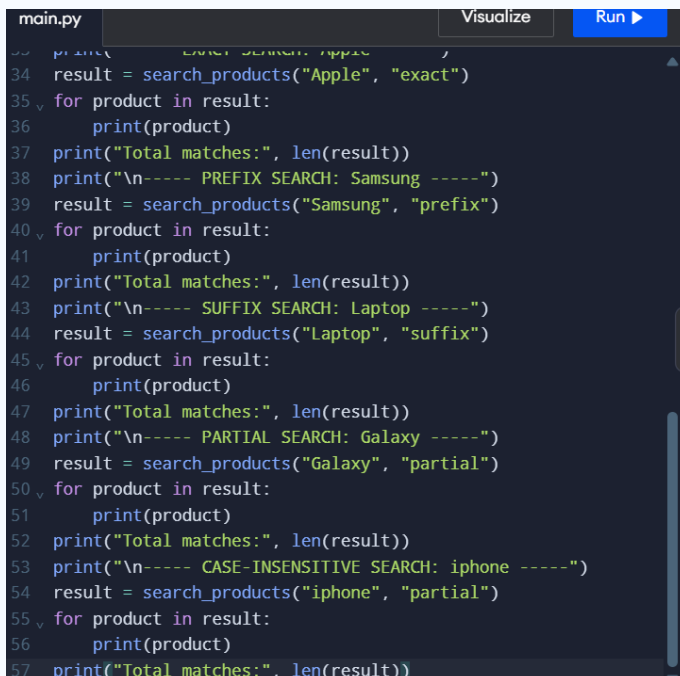
```
Apple MacBook Air
Total matches: 3

----- PREFIX SEARCH: Samsung -----
Samsung Galaxy S24
Samsung Galaxy A55
Samsung Tablet
Total matches: 3

----- SUFFIX SEARCH: Laptop -----
Dell Laptop
HP Laptop
Total matches: 2

----- PARTIAL SEARCH: Galaxy -----
Samsung Galaxy S24
Samsung Galaxy A55
Total matches: 2

----- CASE-INSENSITIVE SEARCH: iphone -----
Apple iPhone 15
Apple iPhone 15 Pro
Total matches: 2
```



The screenshot shows a Python IDE with a file named `main.py`. The code calls the `search_products` function for various keywords and search types, printing the results and the total number of matches for each search.

```
main.py Visualize Run ▶
```

```
34 print("\nEnter search: Apple")
35 result = search_products("Apple", "exact")
36 for product in result:
37     print(product)
38 print("Total matches:", len(result))
39 print("\n----- PREFIX SEARCH: Samsung -----")
40 result = search_products("Samsung", "prefix")
41 for product in result:
42     print(product)
43 print("Total matches:", len(result))
44 print("\n----- SUFFIX SEARCH: Laptop -----")
45 result = search_products("Laptop", "suffix")
46 for product in result:
47     print(product)
48 print("Total matches:", len(result))
49 print("\n----- PARTIAL SEARCH: Galaxy -----")
50 result = search_products("Galaxy", "partial")
51 for product in result:
52     print(product)
53 print("Total matches:", len(result))
54 print("\n----- CASE-INSENSITIVE SEARCH: iphone -----")
55 result = search_products("iphone", "partial")
56 for product in result:
57     print(product)
58 print("Total matches:", len(result))
```

Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

- Validate the student register number.
- Validate the institutional email address.
- Validate the course code.
- Validate the semester information.
- Validate the student's mobile number.
- Display appropriate validation messages for each field.
- Generate a final registration status report indicating whether the registration is successful.

```
main.py Visualize Run ▶ Output
```

```
1 import re
2
3 register_no = "24CSE1234"
4 email = "24cse1234@sse.simats.edu.in"
5 course_code = "DSA03"
6 semester = "6"
7 mobile = "9876543210"
8 print("==== UNIVERSITY REGISTRATION SYSTEM ====")
9 if re.fullmatch(r"\d{2}[A-Z]{3}\d{4}", register_no):
10     print("Register Number :", register_no, "-> Valid")
11     reg_valid = True
12 else:
13     print("Register Number :", register_no, "-> Invalid")
14     reg_valid = False
15 if re.fullmatch(r"[\w.-]+@sse\.simats\.edu\.in", email):
16     print("Institutional Email :", email, "-> Valid")
17     email_valid = True
18 else:
19     print("Institutional Email :", email, "-> Invalid")
20     email_valid = False
21 if re.fullmatch(r"[A-Z]{3}\d{2}", course_code):
22     print("Course Code :", course_code, "-> Valid")
23     course_valid = True
24 else:
25     print("Course Code :", course_code, "-> Invalid")
26
27 if reg_valid and email_valid and course_valid and semester_valid:
28     print("\nRegistration Status : SUCCESSFUL")
29     print("All details are valid.")
30 else:
31     print("\nRegistration Status : FAILED")
32     print("Please correct the invalid details.")
```

```
==== UNIVERSITY REGISTRATION SYSTEM ====
Register Number : 24CSE1234 -> Valid
Institutional Email : 24cse1234@sse.simats.edu.in -> Valid
Course Code : DSA03 -> Valid
Semester : 6 -> Valid
Mobile Number : 9876543210 -> Valid

Registration Status : SUCCESSFUL
All details are valid.
```

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context	Application of Concepts	Problem-Solving Approach	Accuracy of Solution	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____